→ Types Of Data:

Structured Data:

Structured data is the data whose elements are addressable for effective analysis. The data is organized into a formatted repository that is typically a database. Ex: Relational data.

Semi-Structured Data:

It is the data that doesn't reside in relational database but has some organizational properties that make it easier to analyse. Ex: XML data.

Unstructured Data:

It is the data which is not organized in a predefined manner or doesn't have a predefined data model, thus not a good fit for a mainstream relational database.
Ex: Word, pdf, text etc.

→ Statistical analysis : Using statics on samples to make decisions about a population

Population

A population is the **entire collection of objects or outcomes** about which information is sought.

→ Types of Population:

1) Tangible:

Populations where the **members are physical objects**, such as cars, bolts, apples, etc., are called **tangible or concrete populations**.

Such populations are assumed to be **always finite** and therefore **involves counting**.

After an item is sampled, the population size decreases by 1.

In principle, one could in some cases return the sampled item to the population, with a chance to sample it again, but this is rarely done in practice.

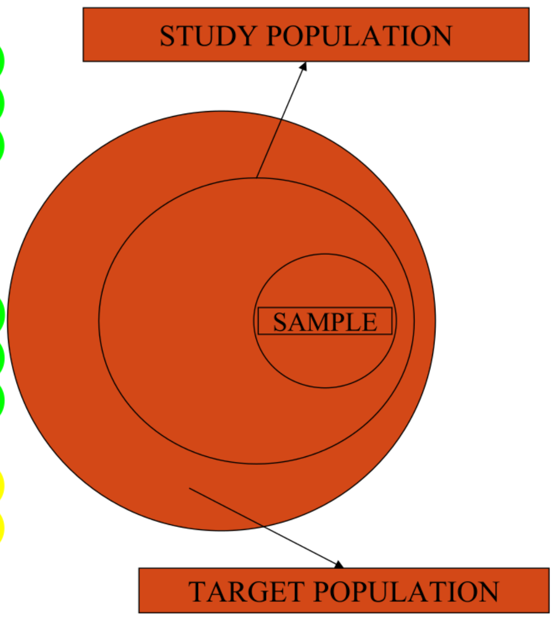
2) Conceptual population:

Populations that **do not consist of physical or actual objects are** called **Conceptual populations**.
Conceptual populations are mostly the **result of a measurement**.
It involves **measuring something multiple times**.
Ex: length of a metal rod.

It consists of a **not well-defined group** of which all elements are not available at the time the sample is collected as **the population increases every day**.
The **size** of a conceptual population is **usually large**.
Ex:a measuring scale population can be all the possible outputs it can give. i.e. infinite. The measured values can be thought of as a sample from this infinite population.

→ Target vs Study Population:

- **Target** or **Theoretical population** refers to the entire group of individuals or objects to which researchers are interested in generalizing the conclusions.
  It must meet a set of criteria of interest to the researchers.
- **Study population** or **accessible population** is the population to which the researches can apply their conclusions to.
  It is a **subset** of the target population. It may be limited to region, state, city, county, or institution



Target Population	Study Population
All institutionalized elderly with Alzheimer's	All institutionalized elderly with Alzheimer's in St Louis county nursing homes
All people with AIDS	All people with AIDS in the metropolitan St. Loui area
All low birth weight infants	All low birth weight infants admitted to the neonatal ICUs in St. Louis city & county
All school-age children with asthma	All school-age children with asthma treated in pediatric asthma clinics in university-affiliated medical centers in the Midwest

→ Terminologies related to Sampling

MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Terminologies related to Sampling

- **Target or Theoretical Population**: The population to which the investigator wants to generalize his results.
- **Sampling Frame** : The sampling frame is the list from which the potential respondents are drawn.
  Ex: List of Universities, List of Students, List of Airline Companies, Telephone Directory
- *****Sampling Unit** : Smallest Unit from which sample can be selected.
- **Sampling Scheme**: Method of selecting sampling units from sampling frame.
- **Sample**: All selected respondents form a sample.ing frame

Sampling Population: Part of Population we have acess to.

* "one Person", "one regiment in the army"
"one Squadron of IAF Jets". The cannot be split / de-constrated further.

→ Characteristics of a sample

Characteristics of a sample

- A sample must be **representative of the population**.
- It must be **appropriately sized**. i.e. it must be sufficiently large to represent the population and provide statistical stability or reliability.
- It must be **unbiased**. It should contain **all types of groups/units** present in the population in **fair proportions**.
- It must be **selected at random**. This means that any item in the group has an equal chance of being and selected and included in the sample.
- It must be **economical**. The objectives of the survey must be achieved in as minimum of cost and effort as possible.
- It must be **goal-oriented**. It must be oriented to the research objectives and fitted to the survey conditions.

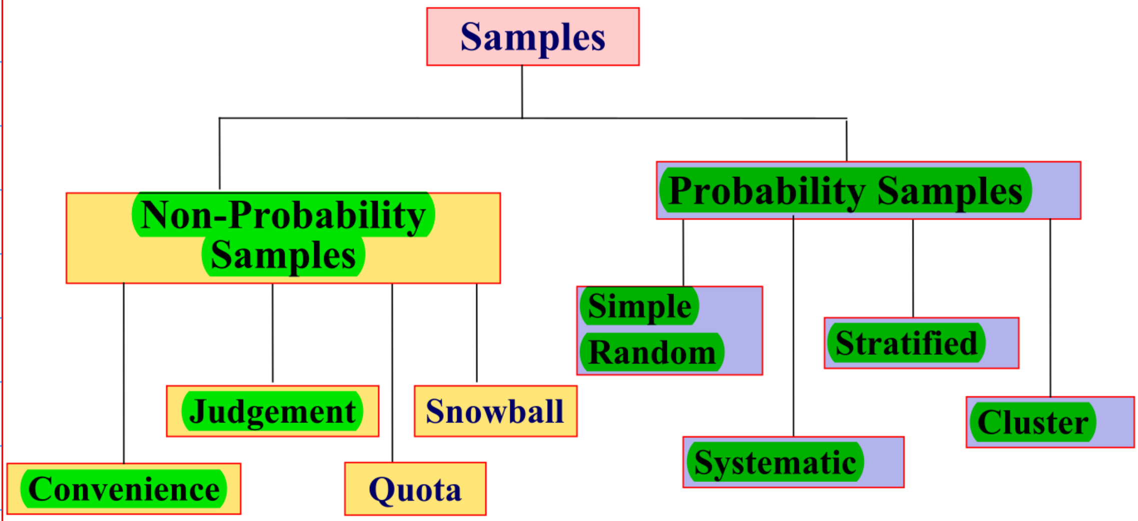
→ Factors that influence sample representativeness:

- Sampling procedure
- Sample size
- Participation (response)

→ When might you sample the entire population?

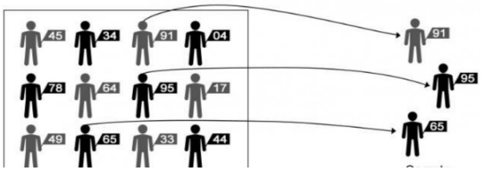
- When your population is very small
- When you have extensive resources
- When you don't expect a very high response

Types of Sampling methods



Probability Sampling

- **Probability sampling** is a type of sampling in which every unit in the population has a chance/probability (greater than zero) of being selected in the sample, and this probability can be accurately determined.
- This type of sampling decreases bias and sampling error in the selection process.
- When every element in the population does have the same probability of selection, this is known as an **'equal probability of selection' (EPS)** design. Such designs are also referred to as **'self-weighting'** because all sampled units are given the same weight.



- **Non-Probability sampling** is a type of sampling in which every unit in the population doesn't have a chance/probability (greater than zero) of being selected in the sample.
- Here, some elements of the population have no chance of selection (these are sometimes referred to as **'out of coverage'/'undercovered'**), or the probability of selection can't be accurately determined.
- It involves the **selection** of elements **based on assumptions regarding the population of interest**, which forms the criteria for selection.
- The **selection** of elements is **non random**.
- Thus, non-probability sampling does not allow the estimation of sampling errors.
- It is more likely to produce a **biased sample** and **restricts generalization.**
- It is not an appropriate data collection method for most of the statistical analysis.

→ Sampling Method:

I. Fundamental Concepts: Samples, Bias, and Error

Sampling involves selecting a subset of units from a larger **Population**. The goal is to obtain a **Representative Sample**, which accurately reflects the characteristics of the population, thereby avoiding a **Biased Sample**.

The entire subject matter is divided into two primary categories: **Probability Samples** and **Non-Probability Samples**.

Errors in Sampling are divided into two main types:

1. **Sampling Error (Random error):** This is the discrepancy between a sample statistic and its population parameter. It occurs when the sample is simply **not representative of the population**. This error is rooted in **sampling variation**, where two different samples from the same population will naturally differ from each other.
2. **Non-sampling Error (Systematic error):** These are mistakes made during data collection or processing (e.g., data entry errors, defective questionnaires, or failure to locate the correct household).

Sampling Bias is distinct from error, occurring when the sampling technique causes the chosen sample to be non-representative. Major forms of bias include **selection bias** (some population members have a higher or lower sampling probability than others) and **non-response bias** (resulting from the absence of subjects who refuse to respond or cannot be contacted).

→ Probabbility Sampling:

II. Probability Sampling Techniques

Probability sampling ensures that **every unit in the population has a probability (greater than zero) of being selected**, and this probability can be accurately determined. This type of sampling inherently decreases bias and sampling error. A key design principle is the **Equal Probability of Selection (EPS) design**, where every element has the same chance of selection, making the design 'self-weighting'.

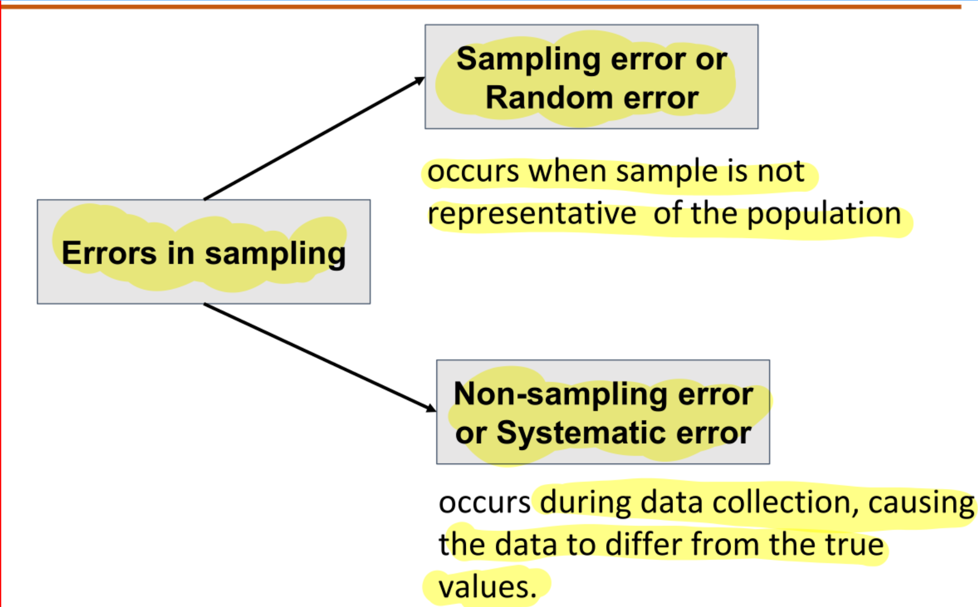
Method	Key Mechanism & Use	Advantages	Disadvantages
Simple Random Sampling (SRS)	**Entirely random**; each unit has an equal chance of selection. Requires a complete **sampling frame**. Best used when the population is **small**.	**Simple to use**; estimates are easy to calculate; usually representative; low sampling error.	Impracticable if the sampling frame is large; **minority subgroups** may not be sufficiently represented; lacks the use of available knowledge concerning the population.
Systematic Sampling	Relies on arranging the population and selecting elements at **regular intervals (k)**. k is the selection interval, calculated as Population Size (N) / Sample Size (n). The **first element is selected randomly**.	Sample is **evenly spread** over the entire population; cost-effective; easy to select.	**Might lead to bias** if there is an **underlying pattern/periodicity** in the population that coincides with the selection interval.
Stratified Sampling	Population is divided into **non-overlapping, internally homogeneous groups (strata)** based on shared characteristics (e.g., age, gender). **Simple random samples are drawn from *each* stratum**. Used when population proportion must be reflected in the sample.	**Enhances representativeness**; ensures adequate representation of minority subgroups; higher statistical efficiency and precision, often requiring a **smaller sample size**.	Sampling frame must be prepared separately for each stratum; **time consuming and expensive**; complex designs can lead to classification errors.
Cluster Sampling	Population is divided into **non-overlapping clusters (areas/groups)**, where each cluster is a miniature of the population (members are **heterogeneous**). **Entire clusters are randomly selected**.	Convenient for geographically dispersed populations; reduces travel costs and simplifies administration; **requires fewer resources**.	**Statistically less efficient** when cluster elements are similar; prone to **higher sampling error and biases** if clusters represent a biased opinion of the whole population.

→ Non - Probability sampling

Non-Probability sampling is characterized by the fact that **every unit does not have a chance/probability of being selected** (or the probability cannot be accurately determined). Selection is **non-random** and based on assumptions or convenience. This method is **more likely to produce a biased sample** and restricts generalization; it also does not allow for the estimation of sampling errors.

Method	Key Mechanism & Use	Advantages	Disadvantages
Convenience Sampling	Also known as grab or opportunity sampling. Sample elements are selected simply because they are **readily available and convenient** to the researcher.	Useful in pilot studies; **low cost and inexpensive** way to gather initial data; simple to implement.	**Prone to significant bias**; sample may not be representative of the population, limiting generalizability and **external validity**.
Judgemental/Purposive Sampling	The researcher **chooses the sample** based on who they think is appropriate, relying on their expertise or knowledge. Used when there is a **limited number of experts** in the area being researched.	Consumes minimum time; allows the researcher to use their expertise.	**Low level of reliability and high levels of bias**; prone to errors in researcher judgment; inability to generalize findings.
Quota Sampling	Population is segmented into subgroups (like stratified sampling), but units are selected based on predetermined characteristics until **pre-defined quota controls are satisfied**. **Judgment, not random selection**, is used for final selection.	Cost-effective; improves representation of certain groups within the population.	**Impossible to determine sampling error**; risk of sampling bias based on ease of access; cannot make statistical inferences leading to problems of generalization.
Snowball Sampling	Subjects are selected based on **referral from initial respondents**. Effective when the desired sample characteristic is **rare** or the sampling frame is difficult to identify (e.g., illegal immigrants or shelterless people).	Allows reaching difficult-to-sample populations via referral; cheap, simple, and cost-efficient.	Significant risk of **selection bias** (referred individuals often share common traits); the researcher has **little control** over the method; representativeness is not guaranteed.

→ Errors in sampling :



MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Sampling error

- The discrepancy between a sample statistic and its population parameter is called sampling error.
- Defining and measuring sampling error is a large part of inferential statistics.
- It occurs when the sample is not representative of the population.
- The sampling error for a given sample is unknown but when the sampling is random, for some estimates (for example, sample mean, sample proportion) theoretical methods may be used to measure the extent of the variation caused by sampling error.

Non-sampling error

- Non-sampling errors are the results of mistakes made in implementing data collection and data processing, such as
  - failure to locate and interview the correct household
  - errors in understanding of the questions by either the interviewer or the respondent
  - data entry errors
  - missing Data
  - poorly conceived concepts, unclear definitions, and defective questionnaires
  - response errors occurring when people are unaware, refuse to answer, or overstate in their answers
- Major sources : Sampling Bias, Non-response Bias.

Sampling Bias

- Sampling bias occurs when a chosen sample is not representative of the larger population.
- It occurs due to the sampling technique/method used to perform data collection.
- It can be either **selection bias and non-response bias.**
- A sampling method has a sampling bias if all subjects in the population are not equally likely to be included in a sample.
- That is, a sample is collected in such a way that some members of the intended population have a lower or higher sampling probability than others.

Selection Bias & Nonresponse bias

➔ **Selection bias:**

- It is a bias in which a sample is collected in such a way that some members of the intended population have a lower or higher sampling probability than others.
- It results in a biased sample of a population in which all individuals, or instances, were not equally likely to have been selected.

➔ **Nonresponse bias:**

- Nonresponse bias is a type of sampling bias that occurs because of the absence of certain objects or subjects from a sample.
- For example, some subjects don't respond to surveys because they refuse, cannot be contacted, or have a lack of interest in the survey content.

→ Sampling variation :

- Simple random samples always differ from their populations in some ways, and occasionally may be substantially different.
- Two different samples from the same population will differ from each other as well.
- This phenomenon is known as **sampling variation**.
- Sampling variation is one of the reasons that scientific experiments produce somewhat different results when repeated, even when the conditions appear to be identical.

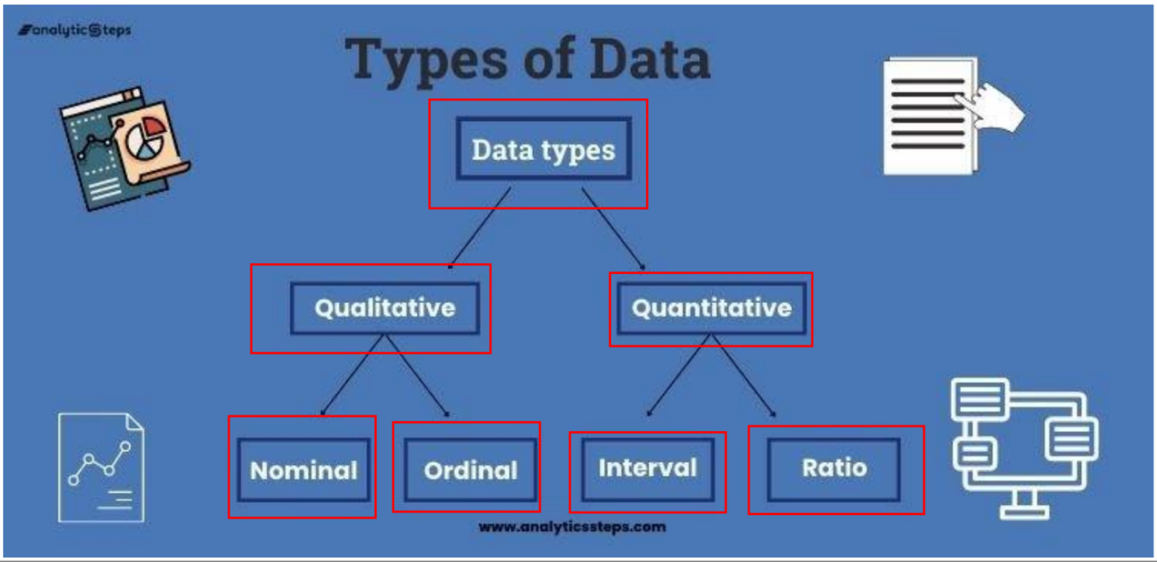
→ Independent Sample :

Independence

- The items in a sample are said to be independent if knowing the values of some of them does not help to predict the values of the others.
- With a finite, tangible population, the items in a simple random sample are not strictly independent, because as each item is drawn, the population changes.
- This change can be substantial when the population is small.
- However, when the population is very large, this change is negligible and the items can be treated as if they were independent
- The sample can be considered independent if sample size is smaller than 5% of population size.
- Since conceptual population have infinite/very large size the sample obtained (ex: measuring a rock) is always independent

→ Types of Data :

Types of data



1. Quantitative Data

Quantitative Data consists of measurements recorded on a **naturally occurring numerical scale**. This data is easily manipulated statistically and answers questions such as **"how many," "how much," and "how often"**.

General Examples: Age, GPA, Salary, Scores of tests and exams, and weight of a person.

Sub-Type / Scale	Core Characteristic	Key Feature & Description	Examples
Discrete Data	Values are **countable** and separate.	Takes on only certain values by finite 'jumps' (no intermediate values). The number of possible values is finite or limited. Bar charts can display this data.	Number of students in a class, number of defective items in a lot, shoe size, number of oranges in a bag.
Continuous Data	Values can take on **any numerical value** within a finite or infinite interval.	Represents information that can be meaningfully divided into finer levels. It can have an unlimited number of values between the lowest and highest points of measurement.	Height, weight, speed of cars, temperature, time required to complete a project.
Interval Data	Measured along a scale of **equal-sized units**.	Has **no true zero-point** or meaningful zero. Allows ranking items and comparing the *magnitude of differences* between them.	Temperature in **Celsius (°C) or Fahrenheit (°F)** (but not Kelvin), calendar dates.
Ratio Data	Represents the **highest level of measurement**.	Has an **inherent or true zero-point**. Allows formation of quotients in addition to differences. Numerical relationships are meaningful (e.g., 10K is twice as high as 5K).	Age, speed, weight, length, counts, **temperature in Kelvin**, monetary quantities.

Summary of Interval and Ratio Attributes' Mathematical Properties:

Attribute Type	Properties Possessed	Allowed Operations
Interval	Distinctness (=, ≠), Order (<, >), Addition (+, −)	Central measures (mean, median, mode), range, spread (standard deviation).
Ratio	**All 4 Properties** (Distinctness, Order, Addition, Multiplication (*, /))	All mathematical operations are possible; most precise data type.

2. Qualitative Data (Categorical Data)

Qualitative Data consists of measurements that **cannot be recorded on a natural numerical scale** but are instead recorded in categories. Arithmetic operations cannot be applied to these variables. This data answers questions like **"how this has happened," or "why this has happened"**.

General Examples: Year in school, Major, Gender, colors.

Sub-Type	Core Characteristic	Key Feature & Description	Examples
Nominal Data	Used purely for **labeling, naming, or categorizing** variables.	The categories have **no particular order or direction**. It is the least powerful in measurement, having no arithmetic origin or order.	Gender (Women, Men), Hair color, Marital status, College major, method of payment, or any alphanumeric/numeric code without basic order or ranking.
Ordinal Data	Values are **ordered** or ranked.	The order matters, but the **distance or difference between values is not known** or meaningful. It allows for setting up inequalities.	Ranking of users in a competition (first, second, third), economic status (low, medium, high), rating of a product on a scale of 1–10, survey questions like a Likert scale.

Summary of Nominal and Ordinal Attributes' Mathematical Properties:

Attribute Type	Properties Possessed	Analysis Methods
Nominal	Distinctness (=, ≠) only.	Grouping method to calculate frequency or percentage. Hypothesis testing often uses nonparametric tests, such as the Chi-Square test.
Ordinal	Distinctness and Order (<, >).	Arithmetic operations cannot be done because they only show the sequence.

→ Types of Studies:

We do studies to gather information and draw conclusions. The type of conclusion we draw depends on the study method used:

I. **Observational study:** In an observational study, we measure or survey members of a sample without trying to affect them.
II. **Controlled study:** In a controlled experiment, we assign people or things to groups and apply some treatment to one of the groups, while the other group does not receive the treatment.

→ Statistics

→ Sample Statistic vs Population Parameter:

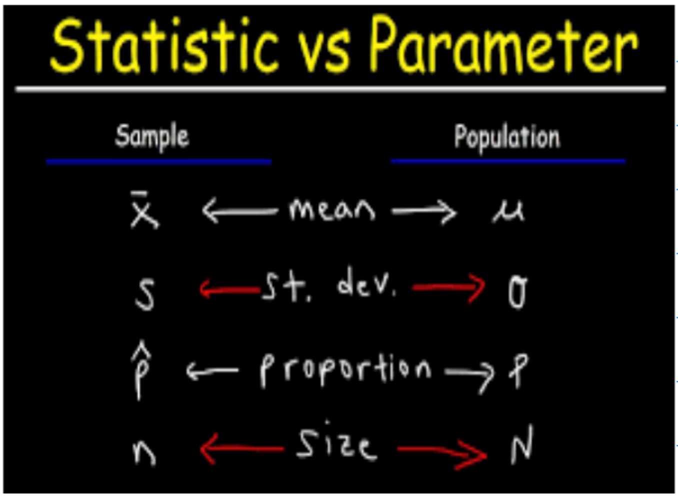
Sample Statistic and Population Parameter

→Sample statistic:

- It is a numerical measurement describing some **characteristic** of a **sample.**
- Example: sample average, median, sample standard deviation, and percentiles.

→Population parameter:

- It is a numerical measurement describing some **characteristic** of a **population**.
- Example: mean and variance of population are population parameters.



→ Descriptive Statistics

Descriptive Statistics vs Inferential Statistics

S. No	Descriptive Statistics	Inferential Statistics
1	Concerned with the describing the target population	Make inferences from the sample and generalize them to the population.
2	Organize, analyze and present the data in a meaningful manner	Compares, test and predicts future outcomes.
3	Final results are shown in form of charts, tables and Graphs	Final result is the probability scores.
4	Describes the data which is already known	Tries to make conclusions about the population that is beyond the data available.
5	Tools- Measures of central tendency (mean/median/ mode), Spread of data (range, standard deviation etc.)	Tools- hypothesis tests, Analysis of variance etc.

→ Measure of Central tendency :

There are three different types of 'average'.
These are the mean, the median and the mode.
They are used by statisticians as a way of summarizing where the 'centre' of the data is.

$$\text{Mean} = \frac{\text{sum of all values}}{\text{total number of values}}$$

Median = middle value (when the data are arranged in order)

Mode = most common value

1) Mean :

- Mean is the **arithmetic average** computed by summing all the values in the dataset and dividing the sum by the number of data values.
- The population mean is represented by Greek letter μ.
- For a finite set of dataset with measurement values X1, X2, ..., Xn (a set of n numbers), it is defined by the formula:

a) Population mean: $\mu_x = \sum_{i=1}^{N} \frac{x_i}{N} = \frac{x_1 + x_2 + \ldots + x_N}{N}$ where N is the population Size

a) Sample mean: $\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$ where n is the sample size

Slide Courtesy:Dr.Uma

- Weighted mean is an average where certain values of the data set contribute more to the mean value.
- For a finite set of dataset with measurement values X1, X2, ..., Xn

(a set of n numbers), and the corresponding weights w1, w2,....wn

it is defined by the formula:

$$\bar{x} = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}$$

mean PDF : $\bar{X} = \sum x_i P_i$

mean CMF : $\bar{x} = \int x P$

- If p% of the data are trimmed from each end, the resulting trimmed mean is called the **"p% trimmed mean"**.
- There are no hard-and-fast rules on how many values to trim.
- The most commonly used trimmed means are the 5%, 10%, and 20% trimmed means.

If the sample size is denoted by n, and a p% trimmed mean is desired, the number of data points to be trimmed is np/100
It is used to reduce the effects of outliers on the calculated average.
This method is best suited for data with large, erratic deviations or extremely skewed distributions.

2) Median :

Median is the value separating the higher half from the lower half of a data sample, a population, or a probability distribution.
It is the middle number in a sorted, ascending or descending, list of numbers and can be more descriptive of that data set than the average.
For a data set, it may be thought of as "the middle" value.
The basic feature of the median in describing data compared to the mean (often simply described as the "average") is that it is not affected by a small proportion of extremely large or small values, and therefore provides a better representation of a "typical" value.

➔ Process of calculating median:

1. Arrange all the values of the data set in ascending order.
   X1,X2,X3,....,Xn
1. Find the **middle position**.
3. The element corresponding to middle position is considered as median if odd number of elements are present.
   i.e. if n is odd, median = (n+1/2)th element's value
4. If there are even number of elements present then the average of the elements present in the middle positions is considered as median. i.e. if n is even,

median = ((n/2)th element's value + (n/2 + 1)th element's value))/2

3> Mode :

- The **mode** is the value that appears most often in a set of data values
- Like the statistical mean and median, the mode is a way of expressing, in a (usually) single number.
- To calculate the mode, we need to look at which **value appears the most often**.
- Example, the mode of the sample [1, 3, 6, 6, 6, 6, 7, 7, 12, 12, 17] is 6.
- Given the list of data [1, 1, 2, 4, 4] its mode is not unique.
  It has 2 modes: 1 and 4
- A dataset, in such a case, is said to be **bimodal**, while a set with more than two modes may be described as **multimodal**.
- **Empirical formula:**

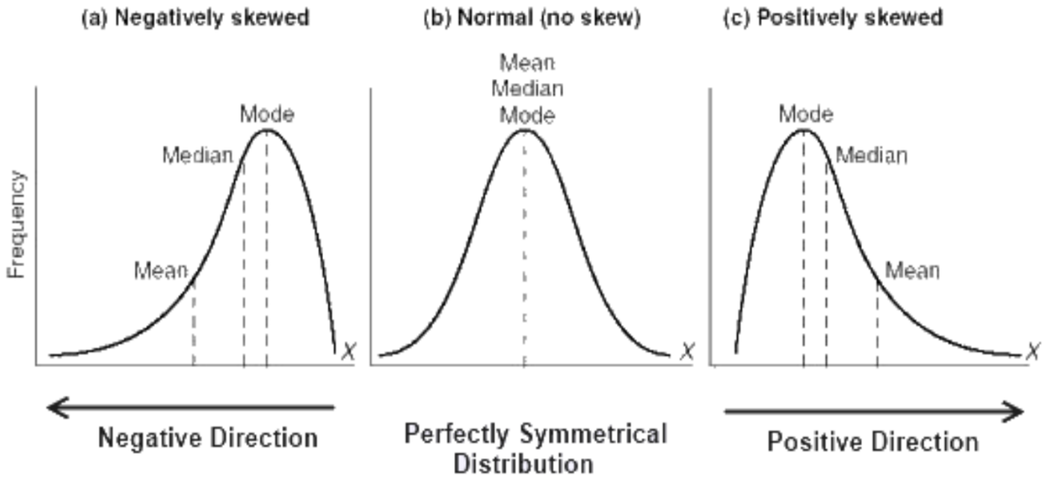
$$\boxed{mean - mode = 3 \times (mean - median)}$$

$$Mode = 3(median) - 2(mean)$$

→ Types of distributions :

- **Skewness** is a **measure of the asymmetry** of the distribution of about its mean.
- The skewness value can be positive, zero, negative, or undefined.
- **Symmetric Distribution:** A symmetric distribution is one where the left and right hand sides of the distribution are roughly equally balanced around the mean.
- In symmetric distributions, the mean, median, and mode are the same.
- **Skewed Distribution:** A skewed distribution is one where the left and right hand sides of the distribution are not balanced around the mean.
- In skewed data, the mean and median lie further toward the skew than the mode.
- The greater the distance of mean and median, the greater is the skewness of the distribution.

Skewed and Symmetric distributions



Left Skewed

Right Skewed

Skewed and Symmetric distributions

- **Distribution of a variable:** tells us what values the variable takes and how often it takes these values.
- **Shape:** It is the "shape" of the distribution of the data.
- If **mean = median = mode**, the shape of the distribution is **symmetric**.
- If **mode < median < mean**, the shape of the distribution trails to the right, is **positively skewed**.
- If **mean < median < mode**, the shape of the distribution trails to the left, is **negatively skewed.**
- Distributions of various "shapes" have different properties and names such as the **"normal" distribution**, which is also known as the **"bell curve"** (among mathematicians it is called the **Gaussian distribution**)

When to use mean, median and mode?

TYPE OF VARIABLE	BEST MEASURE OF CENTRAL TENDENCY
Nominal	Mode
Ordinal	Median
Interval / Ratio (not skewed)	Mean
Interval / Ratio (skewed)	Median

→ Measure of Spread and dispersion

Measures of Spread/Dispersion

- **Absolute Measure of Dispersion:**

  It contains the same unit as the original data set. Absolute dispersion method expresses the variations in terms of the average of deviations of observations like standard or mean deviations. It includes range, standard deviation, quartile deviation, etc.

- **Relative Measure of Dispersion:**

  The relative measures of dispersion are used to compare the distribution of two or more data sets. This measure compares values without units. Common relative dispersion methods include: Coefficient of Range, Coefficient of Variation, Coefficient of Standard Deviation, Coefficient of Quartile Deviation, Coefficient of Mean Deviation.

1) Range :

Measures of Spread: Range

- Range is the most common and easily understandable measure of dispersion.
- It is the difference between two extreme observations of the data set.
- If X max and X min are the two extreme observations then

Range = Xmax – Xmin

5, 13, 9, 7, 1, 9, 2, 9, and 11

↓ put in ascending order

1, 2, 5, 7, 9, 9, 9, 11, 13

↑ ↑

Here, range = 13 - 1

= 12

2) Quartiles:

Measures of Spread: Quantiles

When presenting or analysing measurements of a continuous variable it is sometimes helpful to group subjects into several equal groups.

For example, to create four equal groups we need the values that split the data such that 25% of the observations are in each group.

The cut off points are called quartiles, and there are three of them (the middle one also being called the median).

Likewise, we use two tertiles to split data into three groups, four quintiles to split them into five groups, and so on.

other values likely to be encountered are deciles, which split data into 10 parts, and centiles, which split the data into 100 parts (also called percentiles).

Values such as quartiles can also be expressed as centiles; for example, the lowest quartile is also the 25th centile and the median is the 50th centile.

- A **percentile** is a comparison measure between a particular value and the values of the rest of the data set.
- It shows the percentage of values that a particular element has surpassed.

→ Percentile rank:

The percentile rank is calculated using the formula

R= (P/100)* (N+1) where P is the desired percentile and N is the number of data points.

The pth percentile of a sample, for a number p between 0 and 100, divides the sample such that,

- p% of the sample values are less than the pth percentile
- (100-p%) are greater than the pth percentile

Score vs Percentile

Steps to calculate the percentile rank:

→ Order the n samples values from smallest to largest.

→ Compute the quantity (P/100)(n+1), where n is the sample size.

→ If the above quantity is an integer, the sample value in this position is the percentile.

→ Otherwise, average the two sample values at the preceding and succeeding integer positions with respect to the quantity obtained in step 3.

→ Quartiles :

- Quartiles are the values that **divide a list of numbers into quarters**.
- Quartiles are obtained by first putting the list of numbers in order and then cutting the list into four equal parts.
- The Quartiles are at the "cuts" in the data.
- The first quartile, (Q1) is the middle number between the smallest number and the median of the data.
- The second quartile, (Q2) is the median of the data set.
- The third quartile, (Q3) is the middle number between the median and the largest number.

Measures of Spread: Quartiles

- To find the first quartile, compute the value 0.25(n +1).
- If this is an integer, then the sample value in that position is the first quartile.
- If not, then take the average of the sample values on either side of this value.

$R_{25} = 0.25(n+1)$ i.e. $P = 25$ in "$R = (P/100)(N+1)$"

$R_{50} = 0.50(n+1)$

$R_{75} = 0.75(n+1)$

→ IQR :

- Interquartile range is the distance or range between the 25th percentile and the 75th percentile.
- That is, quantifies the difference between the third and first quartiles.
- **Interquartile Range** = Upper Quartile(Q3) – Lower Quartile(Q1)

IQR = Q3 –Q1

➔ Steps to find IQR :

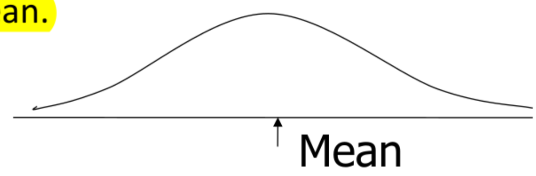
1. Arrange the data scores in ascending order.
2. Find the median of the data set(the number in the middle).
3. Find the median of the lower half of the scores (Q1).
4. Find the median of the upper half of the scores (Q3).

→ ! Dont use Percentile Rank formula here

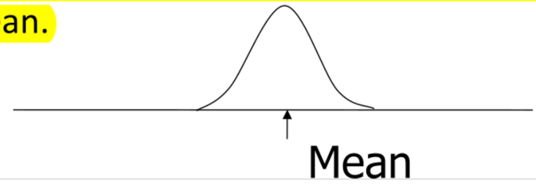
Note: If the number of scores is even, the median is the average of the two middle scores.

3> Variance:

- Variance is a measure of the spread of the recorded values on a variable.
- It is a measure of dispersion, meaning it is a measure of how far a set of numbers is spread out from their average value.
- The larger the variance, the further the individual cases are from the mean.



- The smaller the variance, the closer the individual scores are to the mean.



- It is the average of the distance that each score is from the mean (Squared deviation from the mean)
- Steps to calculate variance:

1. Find the mean value of the given data values.
2. Subtract mean from each data value.
3. Square each value that is obtained from step2.
4. Find the sum of all values that is obtained from step 3.
5. Divide the result that is obtained from step 4 by N(for population) and n-1(for sample).

Variance

$$s^2 = \frac{\Sigma (x - \bar{x})^2}{n - 1}$$ Sample Variance

$$\sigma^2 = \frac{\Sigma (x - \mu)^2}{N}$$ Population Variance

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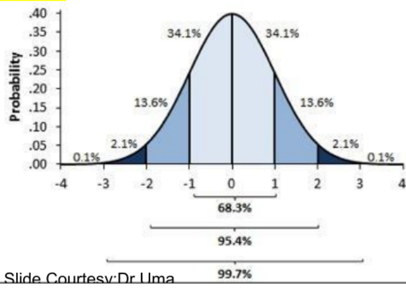
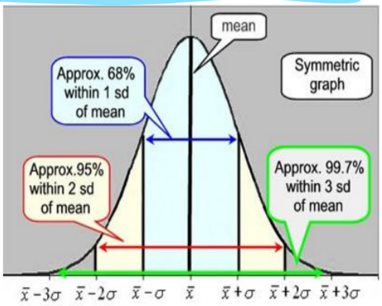
4> Standard deviation:

- Standard deviation signifies the deviation of the elements of the data set from the mean value of the distribution.
- It quantifies the amount of variation of a set of data values.
- It is a measure of the variability of a single item.
- The standard deviation does not decline as the sample size increases.
- The estimate of the standard deviation becomes more stable as the sample size increases.





Source: exceluser.com, MathBitsNotebook

Slide Courtesy:Dr.Uma

- Larger the standard deviation, greater amounts of variation around the mean.
- Std deviation = 0 only when all values are the same (only when you have a constant and not a "variable")
- If you were to "rescale" a variable, the s.d. would change by the same magnitude.
- Like the mean, the standard deviation will be inflated by an outlier case value.

$$S.D = \sqrt{Variance}$$

keep in mind 'S' v/s 'σ'.
Sample v/s Population

→ Chebyshev's Inequality

- The mean of a random variable is a measure of the centre of its distribution, and the standard deviation is a measure of the spread.

- ***Chebyshev's inequality*** *relates the mean and the standard deviation by providing a bound on the probability that a random variable takes on a value that differs from its mean by more than a given multiple of its standard deviation.*

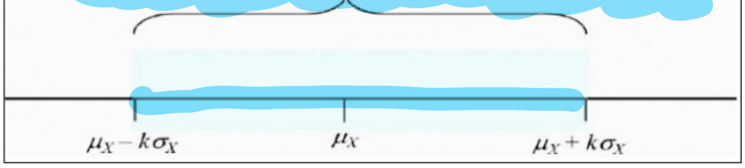
- Specifically, the probability that a random variable differs from its mean by k standard deviations or more is never greater than $1/k^2$.

Chebyshev's inequality states that at least 1 – 1/K^2 of data from a sample must fall within K standard deviations from the mean, where K is any positive real number greater than one.

$$P(|X-\mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Chebyshev's inequality states that at least 1-1/K^2 of data from a sample must fall within K standard deviations from the mean, where K is any positive real number greater than one.

$$P(\mu_X - k\sigma_X < X < \mu_X + k\sigma_X) \geq 1 - \frac{1}{k^2}$$



→ Linear Combination of Random Variable

Addition – Adding a constant to each value of X.

Subtraction – Subtracting a constant from each value of X.

Multiplication – Multiplying each value of X by a constant.

Division – Dividing each value of X by a constant.

where, X represents a Random Variable.

1) Addition:

If X is a random variable and b is a constant, then

$$\boxed{\begin{gathered}\mu_{X+b} = \mu_X + b \\ \sigma^2_{X+b} = \sigma^2_X\end{gathered}}$$

2) multiplication:

Multiplication by a constant affects the mean, variance, and standard deviation of a random variable:

- Mean gets multiplied by the constant

If X is a random variable and a is a constant, then

$$\mu_{aX} = a\mu_X$$

- Variance is multiplied by square of the constant
- Standard Deviation gets multiplied by modulus value of the constant

If X is a random variable and a is a constant, then

$$\sigma^2_{aX} = a^2\sigma^2_X$$

$$\sigma_{aX} = |a|\sigma_X$$

3) multiplying and adding a constant:

Mean: $\mu(aX+b) = a\mu_X + b$ or $E(aX+b) = aE(X) + b$

Variance: $\sigma^2(aX+b) = a^2\sigma^2_X$ or $V(aX+b) = a^2V(X)$

Standard Deviation: $\sigma(aX+b) = |a|\sigma(X)$

Applying Transformations to Random Variable X

Tranformation	Effect on mean	Effect on Variance	Effect on SD	Effect on shape of probability histogram
Add Ex : Y = X + 2	E(Y) = E(X) + 2	Var(Y) = Var(X)	SD(Y) = SD(X)	Doesn't change
Subtract Ex : Y = X - 2	E(Y) = E(X) - 2	Var(Y) = Var(X)	SD(Y) = SD(X)	Doesn't change
Multiplying by a constant Ex: Y = X * 2	E(Y) = E(X) * 2	Var(Y) = Var(X) * 2^2	SD(Y) = SD(X) * 2	Doesn't change
Dividing by a constant Ex: Y = X/2	E(Y) = E(X) / 2	Var(Y) = Var(X) /2^2	SD(Y) = SD(X) / 2	Doesn't change

→ Combing random Variable:

If $X_1, \ldots, X_n$ are random variables and $c_1, \ldots, c_n$ are constants, then the random variable

$$c_1X_1 + \cdots + c_nX_n$$

is called a **linear combination** of $X_1, \ldots, X_n$.

If $X_1, X_2, \ldots, X_n$ are random variables, then the mean of the sum $X_1 + X_2 + \cdots + X_n$ is given by

$$\mu_{X_1+X_2+\cdots+X_n} = \mu_{X_1} + \mu_{X_2} + \cdots + \mu_{X_n} \tag{2.47}$$

If X and Y are random variables, and a and b are constants, then

$$\boxed{\mu_{aX+bY} = \mu_{aX} + \mu_{bY} = a\mu_X + b\mu_Y} \tag{2.48}$$

More generally, if $X_1, X_2, \ldots, X_n$ are random variables and $c_1, c_2, \ldots, c_n$ are constants, then the mean of the linear combination $c_1X_1 + c_2X_2 + \cdots + c_nX_n$ is given by

$$\mu_{c_1X_1+c_2X_2+\cdots+c_nX_n} = \boxed{c_1\mu_{X_1} + c_2\mu_{X_2} + \cdots + c_n\mu_{X_n}} \tag{2.49}$$

If $X_1, X_2, \ldots, X_n$ **are independent, then the** **variance** **of** $Y = c_1X_1 + c_2X_2 + \cdots + c_nX_n$ **is**

$$V(Y) = c_1^2\sigma_1^2 + c_2^2\sigma_2^2 + \cdots + c_n^2\sigma_n^2,$$

When all Constants c1,c2,...,cn = 1 then

$$\sigma^2_{X_1+X_2+\cdots+X_n} = \sigma^2_{X_1} + \sigma^2_{X_2} + \cdots + \sigma^2_{X_n}$$

If X and Y are *independent* random variables with variances σ_X^2 and σ_Y^2, then the variance of the sum $X + Y$ is

$$\sigma^2_{X+Y} = \sigma^2_X + \sigma^2_Y \tag{2.54}$$

The variance of the difference $X - Y$ is

$$\sigma^2_{X-Y} = \sigma^2_X + \sigma^2_Y \tag{2.55}$$

- Here's a few important points about combining random variables:
  - Make sure that the **random variables are independent** or that it's reasonable to assume independence, before combining variances.
  - **Even when we subtract random variables, we still add their variances**; subtracting two variables increases the overall variability in the outcomes.
  - We can find the standard deviation of the combined distributions by taking the square root of the combined variances.

- Means combine very easily with addition or subtraction.

- We can't add standard deviations.

- Variances can be added,

- We can't add standard deviations. square root of variance is the standard deviation.
- Add variances even if subtracting the random variable

→ Mean and Variance of a sample mean

If $X_1, \ldots, X_n$ is a simple random sample from a population with mean μ and variance σ^2, then the sample mean $\overline{X}$ is a random variable with

$$\mu_{\overline{X}} = \mu \tag{2.56}$$

$$\sigma^2_{\overline{X}} = \frac{\sigma^2}{n} \tag{2.57}$$

The standard deviation of $\overline{X}$ is

$$\sigma_{\overline{X}} = \frac{\sigma}{\sqrt{n}} \tag{2.58}$$

→ Central limit theorem

The Central Limit Theorem

Let $X_1, \ldots, X_n$ be a simple random sample from a population with mean μ and variance σ^2.

Let $\overline{X} = \dfrac{X_1 + \cdots + X_n}{n}$ be the sample mean.

Let $S_n = X_1 + \cdots + X_n$ be the sum of the sample observations.

Then if n is sufficiently large,

$$\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{approximately} \tag{4.55}$$

and

$$S_n \sim N(n\mu, n\sigma^2) \quad \text{approximately} \tag{4.56}$$

For most populations, if the sample size is greater than 30, the Central Limit Theorem approximation is good.

✓ If the sampling distribution of $\bar{x}$ is normal or approximately normal, *standardize* or *rescale* the interval of interest in terms of

$$\boxed{z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}}$$

or $Z = \dfrac{\bar{x} - n\mu}{\sqrt{n} * \sigma}$

→ for S_n questions.

✓ Find the appropriate area using Table.

The Sampling Distribution of the Sample Proportion

✓ A random sample of size n is selected from a binomial population with parameter p.

✓ The sampling distribution of the sample proportion,

$$\hat{p} = \frac{x}{n}$$

will have mean p and standard deviation $\sqrt{\dfrac{pq}{n}}$

✓ If n is large, and p is not too close to zero or one, the sampling distribution of $\hat{p}$ will be

The standard deviation of *p*-hat is sometimes called the STANDARD ERROR (SE) of *p*-hat.

✓ If the sampling distribution of $\hat{p}$ is normal or approximately normal, *standardize or rescale* the interval of interest in terms of

$$\boxed{z = \frac{\hat{p} - p}{\sqrt{\dfrac{pq}{n}}}}$$

✓ Find the appropriate area using Z Table .

If $X \sim \text{Bin}(n, p)$, and if $np > 10$ and $n(1-p) > 10$, then

$$X \sim N(np, np(1-p)) \quad \text{approximately}$$

$$\hat{p} \sim N\left(p, \frac{p(1-p)}{n}\right) \quad \text{approximately}$$

→ Random Variable generation for Poission distribution, using acceptence -rejection method.

Acceptance and Rejection Method: Poisson Distribution

Procedure of generating a Poisson random variate N is as follows:

1. Set n=0, P=1
2. Generate a random number R_{n+1}, and replace P by P x R_{n+1}
3. If $P < \exp(-\alpha)$, then accept N=n.
   Otherwise, reject the current n, increase n by one, and return to step 2.